

# ROBERTO HOGAS GORAS

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## Education

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### Purdue University

Exchange program, **Mathematics and Computer Science**

**Aug. 2026 – May 2027**

*West Lafayette, United States*

**Relevant coursework:** CS 555 Cryptography, CS 456 Programming Languages, CS 478 Introduction to Bioinformatics, CS 352 Compilers: Principles and Practice

### University of Florida

Exchange program, **Mathematics and Computer Science**

**Aug. 2025 – Aug. 2026**

*GPA: 3.84/4.00 | Gainesville, United States*

**Relevant coursework:** MAD6407 Numerical Analysis, MAT 4911 Undergraduate Research in Mathematics, CAP5771 Intro to Data Science, CDA4102 Computer Architecture

### University Carlos III of Madrid

Bachelor of Science in **Applied Mathematics and Computer Science**

**Sep. 2023 – May 2027**

*GPA: 3.36/4.00 | Madrid, Spain*

**Relevant coursework:** Complex Analysis, Operating Systems, Probability, Data Structures and Algorithms, Computer Structure

## Research Interest

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**Optimization and Decision-Making:** I am particularly motivated by problems at the intersection of machine learning and operations research, including sequential decision-making methods under uncertainty, robust and stochastic optimization, and modeling techniques like model-order reduction in large-scale problems.

**Scientific Machine Learning:** Interested in developing scalable deep neural network and neural operator architectures to integrate AI-accelerated surrogate models with numerical simulations of large-scale physical systems.

**Decision Trees and Graph Algorithms:** On the theoretical side, I am also keen to explore tree-based optimal classification algorithms to improve decision-tree learning, as well as in the design and analysis of graph algorithms.

**Computational Neuroscience:** I am eager to contribute to research bridging neuroscience and machine learning, using deep neural networks and information theory to characterize neural dynamics. I am also interested in understanding how the brain's connectome shapes information processing in neural circuits from a complex-network perspective.

**Numerical Analysis:** I *genuinely love* numerical analysis. During my junior year of undergraduate at the University of Florida, I took the 6000-level graduate course of numerical analysis, where I received an A, and gave a talk on the subject at the Undergraduate Mathematics Symposium. I am always open to exploring problems involving the application of different numerical methods.

## Preprints and Publications

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**Roberto Hogas Goras**, “Learning Parametric ADCIRC-Like Coastal Shallow-Water Dynamics with Physics-Informed and Operator-Learning Surrogate Models” Unpublished manuscript, 2026. [[pdf](#)] [[code](#)]

## Selected talks

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**Faber Polynomials and Their Applications in Numerical Analysis** [[slides](#)]

**May 2026**

*Undergraduate Mathematics Symposium*, University of Florida, Gainesville, FL

**Learning, Understanding and Modeling for Emerging Materials for Space Photovoltaics (LUMENS-PV)** [[poster](#)]

**Mar. 2026**

*Data Science and Informatics Symposium*, University of Florida, Gainesville, FL

## Experience

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### Undergraduate Research Assistant

Advisor: Dr. Zhe Jiang (University of Florida)

May 2026 – Aug. 2026

Gainesville, United States

- Reviewed relevant literature involving shallow-water equations, physics-informed neural networks, and neural operators for coastal-flow modeling.
- Implemented a numerical solver for the governing shallow-water PDEs to generate synthetic reference datasets across simplified ADCIRC-like coastal scenarios.
- Developed a physics-informed neural network (PINN) to approximate free-surface elevation and depth-averaged velocity fields across varying coastal scenarios.
- Implemented a Deep Operator Network (DeepONet) to learn mappings from physical parameters, forcing conditions, and bathymetry to the corresponding flow solutions generated by the numerical solver.
- Compared both models in terms of predictive accuracy, computational cost, and their ability to generalize to physical scenarios not included during training, presenting the results through interactive demonstrations implemented in Jupyter notebooks.
- Prepared a comprehensive first-author manuscript presenting the theoretical formulation, computational methodology, and experimental results for submission to a machine learning journal.

### Undergraduate Research Assistant

Advisor: Dr. Sara Pollock (University of Florida)

Jan. 2026 – Apr. 2026

Gainesville, United States

- Examined Dr. Pollock's work on accelerating power iteration using Faber polynomial-based momentum methods for matrices whose non-dominant complex eigenvalues lie within a specific deltoid-shaped region.
- Presented the findings at the University of Florida's annual Undergraduate Mathematics Symposium.

### Private Mathematics tutor

University Carlos III of Madrid

Nov. 2024 – Aug. 2025

Madrid, Spain

- Provided individualized mathematics tutoring to high school students, improving their grades and exam performance.
- Gave academic support in preparing for university entrance exams.

### Software Engineer

SAETA UC3M

Oct. 2024 – Aug. 2025

Madrid, Spain

- Designed and implemented an interactive inventory system for the association's website using JavaScript, HTML, Bootstrap, CSS, PostgreSQL, and Django.
- Developed features to add, edit, and delete items, enabling efficient management of association stock.
- Built a private LinkedIn-style platform for the association, enabling companies to filter and search members based on their skills and expertise, using PostgreSQL, Django RESTful APIs and JSON for dynamic data handling.

## Honors & Awards

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- **Best Poster Award**, Data Science and Informatics Symposium, University of Florida. Mar. 2026.
- **Horizonte Global Scholarship**, University Carlos III of Madrid. Selected as one of 13 recipients from more than 1,000 applicants to support with study abroad associated costs. Sep. 2025.

## Technical Skills

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**Programming Languages:** C/C++, Python, PostgreSQL, JavaScript, HTML/CSS

**Technologies:** Git, Docker, Anaconda, Jupyter Notebook

**Libraries:** PyTorch, TensorFlow, NumPy, Pandas, Matplotlib